

Nikhitha Sriram

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Computer Science student with experience in C++, concurrent programming, and database internals through projects involving buffer pool management, page replacement algorithms, and asynchronous disk I/O. Passionate about systems software and performance-oriented engineering.



SKILLS

- Languages: C++, Rust, Python, C, Dart, JavaScript
- Frameworks :Flutter, Node.js, React
- Databases: Firebase Firestore , MongoDB
- Tools & Technologies : Git, Linux, REST APIs, Socket Programming, Selenium, BeautifulSoup , SQL

EDUCATION

- **B-Tech** in Computer Science Engineering - Pes University | Aug 2023 – Aug 2027
CGPA -9.15
- **12th RVPU** - PCMC | 2021 – 2023
Percentage - 95%
- **10th - ICSE** - Carmel School | 2008 - 2021
Percentage - 93%

EXPERIENCE

Akamai Technologies | Software developer Intern | june-july 2026

- Implemented an end-to-end Natural Language to SQL workflow integrating retrieval, context construction, SQL validation, and execution, with support for both a web interface and a Webex bot.
- Designed a multi-stage RAG architecture for enterprise-scale databases by indexing LLM-generated summaries of table schemas, historical SQL queries, and business documentation, enabling efficient context retrieval within LLM token constraints.

Gauntlet | Software developer Intern | june-july 2025

- Developed a static analysis tool to generate Software Bills of Materials (SBOMs) for C, C++, PHP, and C# projects lacking standard package manifests.
- Parsed source files and build artifacts to identify third-party libraries and dependency relationships for accurate SBOM generation.

PROJECTS

Buffer Pool Manager (CMU BusTub) | C++17

- Implemented the storage layer of a relational database, including a concurrent buffer pool manager, page table, and asynchronous disk scheduler.
- Developed an ARC-based page replacement mechanism with ghost lists to adaptively balance recency and frequency of page accesses.
- Ensured correctness under concurrent workloads using shared mutexes, atomic reference counting, and RAIL-based page guards.

Chip-8 Emulator | Rust:

- CHIP-8 emulator written in Rust with graphics and sound support.
- Complete instruction set implementation - All 35 CHIP-8 opcodes along with graphics rendering

Tracking(App dev) | Flutter , Firebase

- Developed a health monitoring app for dialysis patients using Flutter & Firebase.
- Enabled tracking of weight, fluid intake/output, and dialysis bag timings with graphs and pdf's.
- Implemented Firebase Authentication and Firestore for secure data storage.

Web Scrapping

- Developed a web scraper using Selenium to automate product data extraction from Amazon and Flipkart.
- Collected real-time pricing, titles, and availability information for comparative analysis.

AWARDS & COMPETITIONS

CIE(center of innovation and entrepreneurship) ignite

Secured 2nd place in CIE ignite out of 461 teams and 3000 participants

Academic Scholarship

Consistently awarded for high academic performance at PES University.